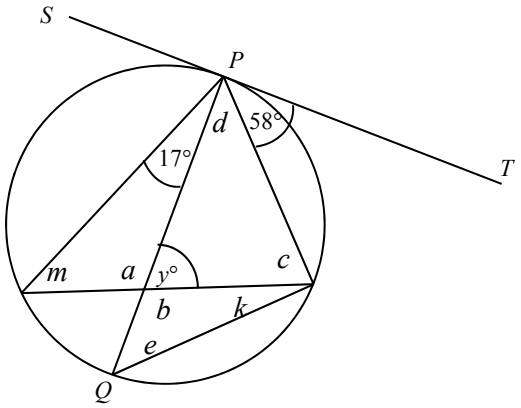
Question	Answer	Marks	Partial Marks
2(b)	75	**5**	**B4** for 105 at *a* or *b* **or** 73 at *c* **and** 32 at *d* or **B3** for 58 at *m* **or** 58 at *e* **and** 17 at *k* or **B2** for 32 at *d* **and** 90 soi at (*c*+*k*) **or** 32 at *d* **and** 17 at *k* **or** 73 at *c* or **B1** for 90 soi at (*c* + *k*) or between tangent and radius **or** 32 at *d* **or** 17 at *k* 
3(a)	$1 - r$	**1**	
3(b)(i)	$(1 - r)(1.3 - r)$ [= 0.4]	**1**	**FT** *their***(a)** dep on **(a)** being an expression in *r*
3(b)(ii)	$1.3 - 1.3r - r + r^2$ or better nfww	**M1**	**FT** *their* **(b)(i)**
	$0.9 - 2.3r + r^2 [= 0]$ OR $13 - 13r - 10r + 10r^2 = 4$ oe	**M1**	**Strict FT** *their* expansion to a quadratic then equating to 0.4 and then collecting to 3 terms on 'one side' OR **Strict FT** *their* expansion to a quadratic = 0.4 all multiplied by 10
	$10r^2 - 23r + 9 = 0$	**A1**	no errors or omissions seen

Question	Answer	Marks	Partial Marks
3(b)(iii)	$(5r-9)(2r-1)$ [= 0]	**B2**	or **B2** for e.g. $5r(2r-1)-9(2r-1)$ **and then** $5r-9=0$ **and** $2r-1=0$ or **B1** for $5r(2r-1)-9(2r-1)$ [= 0] or $2r(5r-9)-1(5r-9)$ [= 0] or $(5r+a)(2r+b)$ [= 0] where a, b are integers and $ab = +9$ or $2a+5b=-23$ If 0 scored, **SC1** for $5r-9$ **and** $2r-1$ seen but not in factorised form
	[r =] $\frac{9}{5}$ oe [r =] $\frac{1}{2}$ oe	**B1**	
3(b)(iv)	0.8 or $\frac{4}{5}$ oe	**1**	
4(a)(i)	1.5 oe	**1**	
4(a)(ii)	(0, 2)	**1**	
4(b)(i)	$y=-2x+6$ oe final answer	**3**	**B2** for $y=-2x+c$ oe or $y=mx+6$ oe $m \neq 0$ or for answer $-2x+6$ or **B1** for [gradient =] $-\frac{6}{3}$ oe or $c = +6$ soi
4(b)(ii)	$y=0.5x-1.5$ oe final answer	**3**	**B1** for [gradient =] – 1 divided by *their* gradient from **(b)(i)** evaluated soi **M1** for substitution of (9, 3) into $y = (\textit{their } m)x + c$ seen in working
4(c)(i)	12.6 or 12.64 to 12.65	**3**	**M2** for $\sqrt{(8--4)^2+(5-1)^2}$ oe or **M1** for $(8--4)^2+(5-1)^2$ oe
4(c)(ii)	(2, 3)	**2**	**B1** for each
5(a)	2.45, 0.25, – 0.25	**3**	**B1** for each
5(b)	Fully correct smooth curve	**4**	**B3FT** for 6 or 7 points or **B2 FT** for 4 or 5 points or **B1 FT** for 2 or 3 points
5(c)	0.7 to 0.8	**1**	**FT** *their* curve
5(d)(i)	Correct ruled line	**2**	**M1** for good freehand, or ruled line with gradient –1.05 to –0.95 or ruled line through (0, 2) but not line $y = 2$